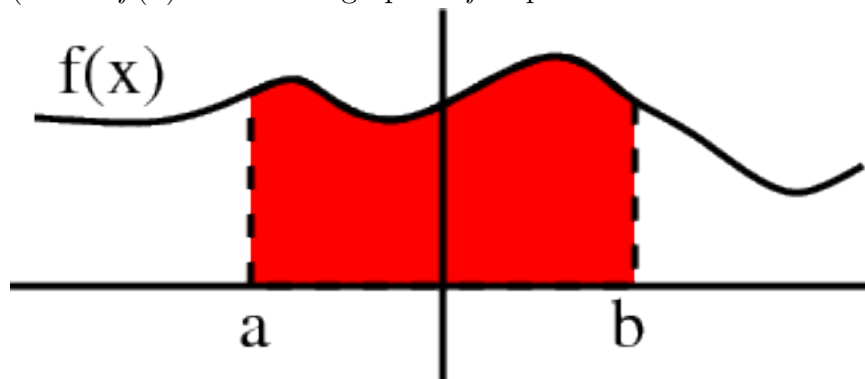


## Module 4: MULTIPLE INTEGRALS - Double Integrals

We know that integration of a function of one variable over an interval will give us the **area** of the function **below the graph of the function between the limits**: that is **area under the curve**: (if  $f(x) > 0$ ).

The interval  $[a, b]$  is called as the **Region of Integration**.

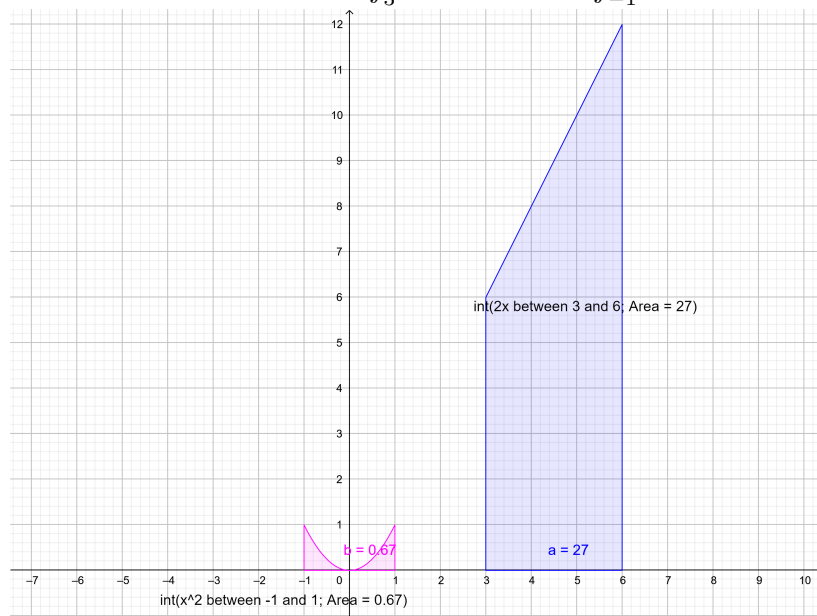
(Note:-  $f(x) > 0$  means graph of  $f$  is present above the  $x$ -axis)



(Ref : [https://mathinsight.org/image/integral\\_area\\_under\\_curve](https://mathinsight.org/image/integral_area_under_curve))

NykampDQ, *Integral as area under a curve*. From MathInsight. <http://mathinsight.org/image/>

For instance, we have :  $\int_3^6 2x \, dx$  and  $\int_{-1}^1 x^2 \, dx$  represented as areas:



In this chapter we consider integrals of functions of two variables  $f(x, y)$  over a region in the plane :  $\int \int_R f(x, y) dx dy$  and the integrals of functions of three variables  $f(x, y, z)$  over a region in space:  $\int \int \int_S f(x, y, z) dx dy dz$

These integrals are called **multiple integrals**.

**Applications of Multiple Integrals:** They are useful in calculating

- areas of regions and volumes of surfaces
- total mass or the angular momentum of an object of varying density
- Average values of functions

Note:- In Double integration:  $f(x)$  gets replaced by  $f(x, y)$  which is a function of two variables  $x$  and  $y$  and the interval  $[a, b]$  gets replaced by some bounded region (most of the time)  $R$  in  $xy$  plane.

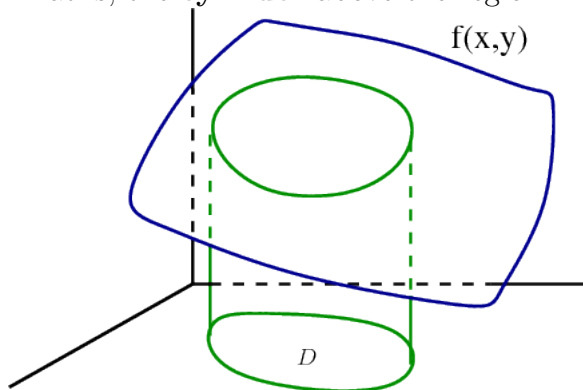
so,  $z = f(x, y)$  represents surface in space and if we are integrating  $f(x, y)$  over the given bounded Region  $R$  then we are finding the volume under the surface  $z = f(x, y)$  over the Region  $R$ .

For example in the following figure  $\int \int_D f(x, y) dA$  can be interpreted as the volume under the surface  $z = f(x, y)$  over the region  $D$ .

Imagine that the blue object below is the surface  $z = f(x, y)$  floating above the  $xy$ - plane.

The double integral  $\int \int_D f(x, y) dA$  can be interpreted as the volume between the surface  $z = f(x, y)$  and the  $xy$ - plane:

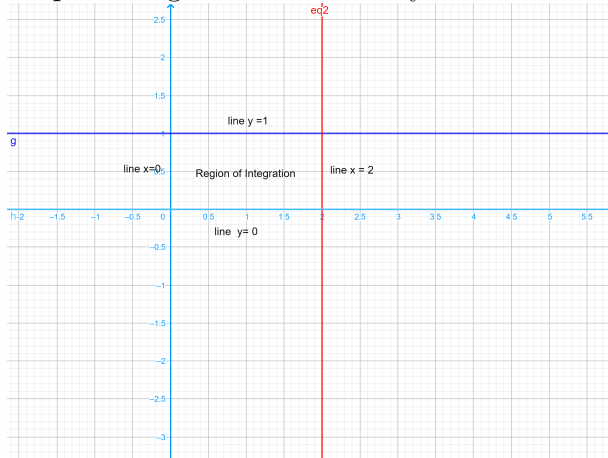
That is, the **cylinder** above the region  $D$ .



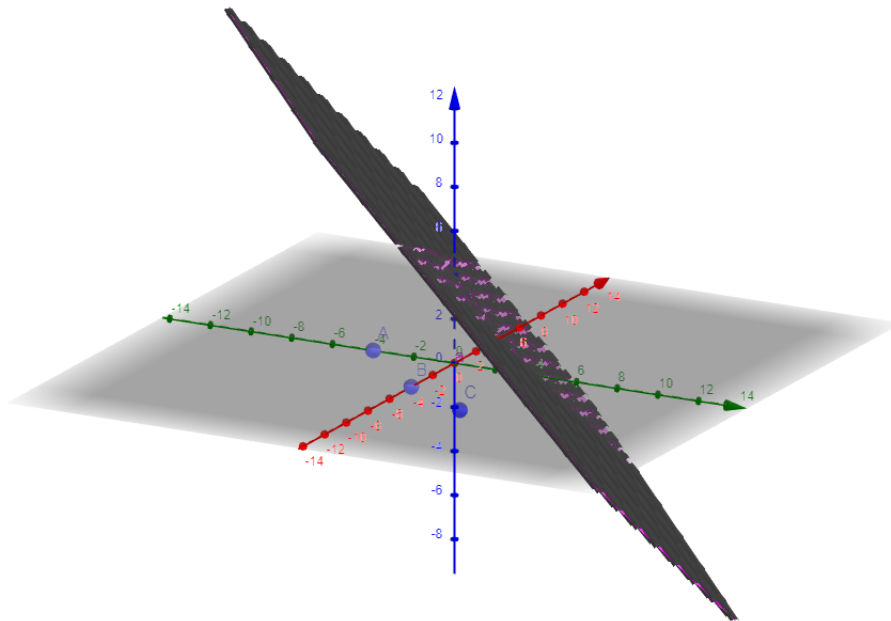
(Ref : [https://mathinsight.org/double\\_integral\\_volume](https://mathinsight.org/double_integral_volume))

For Example:-  $\int_{y=0}^{y=1} \int_{x=0}^{x=2} (4 - x - y) dx dy$

Required region is bounded by the lines  $x = 0, x = 2$  &  $y = 0$  &  $y = 1$



To plot the surface  $z = 4 - x - y$  :



(We see that

|   |   |   |   |   |
|---|---|---|---|---|
| x | 0 | 2 | 2 | 0 |
| y | 0 | 0 | 1 | 1 |
| z | 4 | 2 | 1 | 3 |

)

so by integrating  $z = 4 - x - y$  over the given Region we will get the **volume** under the surface  $z = 4 - x - y$  over the given region.

Let  $R$  be a given Region of Integration depending on the Region  $R$ , Double Integration takes following form.

1. If  $R$  is a Rectangular Region then the Double integration over  $R$  is written as,

$$\begin{aligned}\iint_R f(x, y) dx dy &= \int_{y=c}^{y=d} \int_{x=a}^{x=b} f(x, y) dx dy \\ &= \int_{x=a}^{x=b} \int_{y=c}^{y=d} f(x, y) dy dx\end{aligned}$$

2. If  $R$  is the region bounded by the curves  $x = g_1(y)$  and  $x = g_2(y)$  present between the lines  $y = c$  and  $y = d$  then we have,

$$\iint_R f(x, y) dx dy = \int_{y=c}^{y=d} \int_{x=g_1(y)}^{x=g_2(y)} f(x, y) dx dy$$

OR If  $R$  is the region bounded by the curves  $y = f_1(x)$ ,  $y = f_2(x)$  between the lines  $x = a$  and  $x = b$  then we have,

$$\iint_R f(x, y) dx dy = \int_{x=a}^{x=b} \int_{y=f_1(x)}^{y=f_2(x)} f(x, y) dy dx$$

Note:- In  $\iint_R f(x, y) dx dy$ , limits of inner integral are called as inner limits and the limits of outer integral are called as outer limits.

**Evaluation of Double Integrals**

(when limits are given)

If inner limits are ' $x$ ' limits then integrate with respect to ' $dx$ ' by treating ' $y$ ' as a constant and then finally integrate with respect to ' $dy$ ' and vice versa.

**Solved Examples**

1. Evaluate  $\int_0^1 \int_0^{x^2} e^{y/x} dy dx$

**Solution:** Let

$$\begin{aligned} I &= \int_0^1 \int_0^{x^2} e^{y/x} dy dx \\ &= \int_0^1 \left( \int_0^{x^2} e^{y/x} dy \right) dx \end{aligned}$$

since inner limits are ' $y$ ' limits we will integrate  
w.r.t ' $y$ ' first by treating ' $x$ ' as a constant

$$\begin{aligned} \therefore I &= \int_{x=0}^{x=1} \left[ \frac{e^{y/x}}{\frac{1}{x}} \right]_0^{x^2} dx \\ \therefore I &= \int_{x=0}^{x=1} x [e^{x^2/x} - e^{0/x}] dx \\ \therefore I &= \int_0^1 x (e^x - 1) dx \\ \therefore I &= \int_0^1 x e^x dx - \int_0^1 x dx \\ \therefore I &= [x e^x - e^x]_0^1 - \left[ \frac{x^2}{2} \right]_0^1 \\ \therefore I &= (1 \cdot e^1 - e^1) - (0 \cdot e^0 - e^0) - \left( \frac{1}{2} - 0 \right) \\ \therefore I &= (e - e) - (0 - 1) - \frac{1}{2} \\ \therefore I &= \frac{1}{2} \end{aligned}$$

2. Evaluate  $\int_0^1 \int_{x^2}^x xy(x+y) dy dx$

**Solution:** Let

$$\begin{aligned} I &= \int_0^1 \int_{x^2}^x xy(x+y) dy dx \\ \therefore I &= \int_{x=0}^{x=1} \left( \int_{y=x^2}^{y=x} (x^2y + xy^2) dy \right) dx \\ \therefore I &= \int_{x=0}^{x=1} \left[ \frac{x^2y^2}{2} + \frac{xy^3}{3} \right] \Big|_{y=x^2}^{y=x} dx \\ \therefore I &= \int_{x=0}^{x=1} \left[ x^2 \cdot \frac{x^2}{2} + x \cdot \frac{x^3}{3} - \left( x^2 \cdot \frac{(x^2)^2}{2} + x \cdot \frac{(x^2)^3}{3} \right) \right] dx \\ \therefore I &= \int_0^1 \left[ \frac{x^4}{2} + \frac{x^4}{3} - \frac{x^6}{2} - \frac{x^7}{3} \right] dx \\ &= \int_0^1 \left[ \frac{5}{6} \cdot x^4 - \frac{x^6}{2} - \frac{x^7}{3} \right] dx \\ &= \left[ \frac{5}{6} \cdot \frac{x^5}{5} - \frac{x^7}{14} - \frac{x^8}{24} \right] \Big|_0^1 \\ &= \left( \frac{1}{6} - \frac{1}{14} - \frac{1}{24} \right) - (0) \\ \therefore I &= \frac{3}{56} \end{aligned}$$

3. Evaluate  $\int_0^\infty \int_0^\infty e^{-x^2(1+y^2)} x \, dx \, dy$

**Solution:** Let

$$I = \int_0^\infty \left( \int_0^\infty e^{-x^2(1+y^2)} x \, dx \right) dy$$

since limits of integrations are constant

we integrate with respect to 'x' first

then with respect to 'y'

(i.e do not change the order)

$$\text{let } w = e^{-x^2(1+y^2)}$$

$$\therefore dw = e^{-x^2(1+y^2)} \cdot (-2x)(1+y^2) \, dx$$

$$x = 0, w = 1$$

$$x = \infty, w = 0 \quad (\text{since } e^{-\infty} = 0)$$

Since we are integrating w.r.t 'x'  $\Rightarrow (1+y^2)$  is a constant

$$\therefore I = \frac{1}{2} \int_0^\infty \left( \int_0^1 \frac{-2 \, dw}{2(1+y^2)} \right) dy$$

$$= \frac{1}{2} \int_0^\infty \frac{dy}{(1+y^2)} \left( \int_0^1 dw \right)$$

$$\therefore \int_a^b f(t) \, dt = - \int_b^a f(t) \, dt$$

$$= \frac{1}{2} \int_0^\infty \frac{dy}{(1+y^2)} [1 - 0]$$

$$= \frac{1}{2} [\tan^{-1} y]_0^\infty = \frac{1}{2} \cdot \left( \frac{\pi}{2} - 0 \right)$$

$$\therefore I = \frac{\pi}{4}$$

4. Evaluate  $\int_{-1}^1 \int_0^{1-x} x^{1/3} y^{-1/2} (1-x-y)^{1/2} dy dx$

**Solution:** Let

$$I = \int_{-1}^1 \left( \int_{y=0}^{y=1-x} x^{1/3} y^{-1/2} (1-x-y)^{1/2} dy \right) dx$$

$$\therefore I = \int_{-1}^1 x^{1/3} \left( \int_{y=0}^{y=1-x} y^{-1/2} (1-x-y)^{1/2} dy \right) dx$$

Since 'x' is constant for inner integral to solve the inner integral put  
 $y = (1-x)t \Rightarrow dy = (1-x) dt$   
 $y = 0 \Rightarrow t = 0 \quad \because (1-x)$  is constant for the inner integral and  
 $y = 1-x \Rightarrow t = 1$

$$\begin{aligned} \therefore I &= \int_{-1}^1 x^{1/3} \left( \int_{t=0}^{t=1} [(1-x)t]^{-1/2} ((1-x) - (1-x)t)^{1/2} (1-x) dt \right) dx \\ &= \int_{-1}^1 (1-x)^{-1/2} (1-x)^{1/2} (1-x) x^{1/3} \left( \int_0^1 t^{-1/2} (1-t)^{1/2} dt \right) dx \\ &= \int_{-1}^1 (1-x) x^{1/3} \beta\left(\frac{-1}{2} + 1, \frac{1}{2} + 1\right) dx \\ (\because \int_0^1 x^n (1-x)^m dx &= \beta(n+1, m+1)) \\ \Rightarrow I &= \beta\left(\frac{1}{2}, \frac{3}{2}\right) \int_{-1}^1 (1-x) x^{1/3} dx \\ &= \frac{\Gamma(\frac{1}{2})\Gamma(\frac{3}{2})}{\Gamma(\frac{1}{2} + \frac{3}{2})} \int_{-1}^1 (x^{1/3} - x^{4/3}) dx \\ &= \frac{\sqrt{\pi} \cdot \frac{1}{2} \cdot \sqrt{\pi}}{\Gamma(2)} \left[ \frac{x^{4/3}}{\frac{4}{3}} - \frac{x^{7/3}}{\frac{7}{3}} \right] \Big|_{-1}^1 \\ &= \frac{\pi}{2} \left[ \left( \frac{3}{4} - \frac{3}{4} \right) - \left( \frac{3}{4}(-1)^{4/3} - \frac{3}{7}(-1)^{7/3} \right) \right] \\ &= \frac{\pi}{2} \left[ \frac{3}{4} - \frac{3}{7} - \left( \frac{3}{4} + \frac{3}{7} \right) \right] \\ (since, (-1)^{4/3} &= ((-1)^{1/3})^4 = (-1)^4 = 1 \quad similarly \quad (-1)^{7/3} = -1) \\ \Rightarrow I &= \frac{-3\pi}{7} \end{aligned}$$



5. Evaluate  $\int_0^1 \int_0^{\sqrt{a^2+x^2}} \frac{dy \, dx}{a^2 + x^2 + y^2} \quad (a > 0)$

**Solution:** Let

$$\begin{aligned} I &= \int_{x=0}^{x=1} \left( \int_{y=0}^{y=\sqrt{a^2+x^2}} \frac{dy}{a^2 + x^2 + y^2} \right) dx \\ \Rightarrow I &= \int_0^1 \left( \int_{y=0}^{y=\sqrt{a^2+x^2}} \frac{dy}{(\sqrt{a^2+x^2})^2 + y^2} \right) dx \end{aligned}$$

( since we are integrating w.r.t  $y$ , variable  $x$  is a constant )

$$\begin{aligned} \therefore I &= \int_0^1 \left[ \tan^{-1} \left( \frac{y}{\sqrt{a^2+x^2}} \right) \right] \Big|_{y=0}^{y=\sqrt{a^2+x^2}} \times \frac{1}{\sqrt{a^2+x^2}} dx \\ &= \int_0^1 \left[ \tan^{-1}(1) - \tan^{-1}(0) \right] \cdot \frac{dx}{\sqrt{a^2+x^2}} \\ &= \frac{\pi}{4} \int_0^1 \frac{dx}{\sqrt{a^2+x^2}} \\ &= \frac{\pi}{4} \left[ \log(x + \sqrt{a^2+x^2}) \right] \Big|_0^1 \\ \therefore I &= \frac{\pi}{4} \left[ \log(1 + \sqrt{a^2+1}) - \log(0 + \sqrt{a^2+0}) \right] \\ \Rightarrow I &= \frac{\pi}{4} \log \left( \frac{1 + \sqrt{1+a^2}}{a} \right) \end{aligned}$$

**Evaluation of Integrals over a given region** (when limits are not specifically given)

**To Find limits of Integration**

Suppose given Region  $R$  is bounded by two curves  $\phi_1(x, y) = 0$ , &  $\phi_2(x, y) = 0$ . When faced with Evaluating  $\iint_R f(x, y) dy dx$ , integrating first with respect to  $y$  & then with respect to  $x$ , do the following:

1. Sketch the region of integration and label the bounding curves and get the points of intersection.

2. To find the  $y$ - limits of integration: (inner limits)

Imagine a vertical line  $L$  cutting through  $R$  in the direction of increasing  $R$ .

That is, we draw vertical lines **parallel** to the  $y$ - axis Mark the  $y$ - values where  $L$  enters and leaves.

These are the  $y$ - limits of integration and are usually functions of ' $x$ ' or constants [i.e we can write  $\phi_1(x, y) = 0$  as  $y = f_1(x)$  and  $\phi_2(x, y) = 0$  as  $y = f_2(x)$ ]

3. Find the  $x$ -limits of integration :(outer limits)

Choose  $x$ -limits that include all the vertical lines through 'R'. **Note:**

Outer limits are always constant

$$\therefore \iint_R f(x, y) dy dx = \int_{x=x_1}^{x=x_2} \int_{y=f_1(x)}^{y=f_2(x)} f(x, y) dy dx$$

In order to write inner limits as  $x$ -limits and outer limits as  $y$ -limits:

- (a) Consider horizontal lines instead of vertical lines.

$$\text{Therefore, } \iint_R f(x, y) dx dy = \int_{y=y_1}^{y=y_2} \int_{x=g_1(y)}^{x=g_2(y)} f(x, y) dx dy$$

- (b) If entering boundary or leaving boundary are more than one then we have to split the region of integration.

$R_1$  and  $R_2$  are two parts of the Region  $R$ .

- (c) If  $f(x, y)$  is continuous over region  $R$  then

$$\begin{aligned} \iint_R f(x, y) dx dy &= \int_{x=x_1}^{x=x_2} \int_{y=f_1(x)}^{y=f_2(x)} f(x, y) dy dx \\ &= \int_{y=y_1}^{y=y_2} \int_{x=g_1(y)}^{x=g_2(y)} f(x, y) dx dy \end{aligned}$$

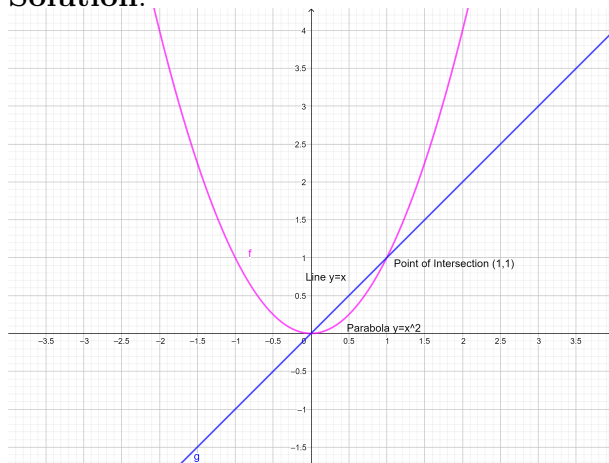
**Note:**  $z = f(x, y)$  is continuous over  $R$  means surface  $z = f(x, y)$  doesn't have breaks

We will apply the above method in solving the following examples:

**Examples:**

1. Evaluate  $\iint_R xy(x+y) dx dy$  where  $R$  is the region bounded by  $y = x^2$  and  $y = x$

**Solution:**



To find the points of intersection:

$$\text{Since, } y = x^2 \text{ and } y = x \Rightarrow x^2 = x \Rightarrow x^2 - x = 0 \\ \Rightarrow x = 0 \text{ or } x = 1$$

if  $x = 0$ ,  $y = 0$  and if  $x = 1$ ,  $y = 1 \therefore$  Points of intersection are  $(0, 0)$  and  $(1, 1)$

To find the limits of integration:-

Since the region  $R$  is bounded by the two curves  $y = x^2$  and  $y = x$  there is no need to split the region.

In order to write the inner limits as  $y$ - limits:

Consider a line parallel to  $y$ - axis (In increasing order of  $y$ ).

So entering boundary is  $y = x^2$  and exit boundary is  $y = x$

Region  $R$  is lying between the lines  $x = 0$  and  $x = 1$

therefore,

$$\iint_R xy(x+y) dx dy = \int_{x=0}^{x=1} \left( \int_{y=x^2}^{y=x} xy(x+y) dy \right) dx = \frac{3}{56}$$

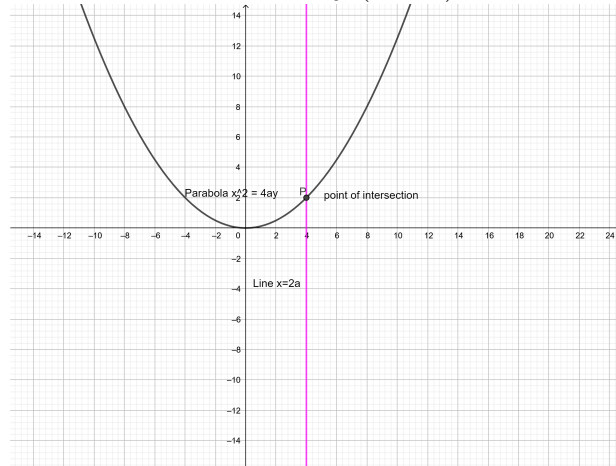
**Note:-** In order to write inner limits as  $x$ - limits we need to consider a line parallel to  $x$ - axis. (In increasing order of  $x$ ).

So entering boundary is  $x = y$  and exit boundary is  $x = \sqrt{y}$ .

(since,  $x^2 = y$ ,  $\therefore x = \pm\sqrt{y}$  and since the region  $R$  is in the 1<sup>st</sup> quadrant:  $x \geq 0$ )

$$\text{Therefore, } \iint_R xy(x+y) dx dy = \int_{y=0}^{y=1} \left( \int_{x=y}^{x=\sqrt{y}} xy(x+y) dx \right) dy = \frac{3}{56}$$

2. Evaluate  $\iint xy dx dy$  over the region bounded by:  
the  $x$ -axis,  $x = 2a$  and  $x^2 = 4ay$  ( $a > 0$ )



**Solution:**

To find points of intersection:-

$x^2 = 4ay$  and  $x = 2a$   $\therefore (2a)^2 = 4ay \Rightarrow 4a^2 = 4ay \Rightarrow y = a$   
and since  $x = 2a$ , the Point of intersection =  $(2a, a)$ . So

$$\begin{aligned} \iint_R xy dx dy &= \int_{x=0}^{x=2a} \left( \int_{y=0}^{y=\frac{x^2}{4a}} xy dy \right) dx \\ &= \int_{x=0}^{x=2a} x \left[ \frac{y^2}{2} \right]_{y=0}^{y=\frac{x^2}{4a}} dx \\ &= \frac{1}{2} \int_{x=0}^{x=2a} x \left[ \left( \frac{x^2}{4a} \right)^2 - 0 \right] dx \\ &= \frac{1}{32a^2} \int_{x=0}^{x=2a} x^5 dx \\ &= \frac{1}{32a^2} \left[ \frac{x^6}{6} \right]_0^{2a} dx \\ &= \frac{2^6 a^6}{(32a^2)6} \\ &= \frac{a^4}{3} \end{aligned}$$

$$\left( \text{ or } \iint_R xy \, dx \, dy = \int_{y=0}^{y=a} \left( \int_{x=\sqrt{4ay}}^{x=2a} xy \, dx \right) dy = \frac{a^4}{3} \right)$$

Note:  $x^2 = 4ay \Rightarrow x = +\sqrt{4ay}$  (since the region is present in the first quadrant )

3. Evaluate  $\iint y \, dx \, dy$  over the area bounded by  $y = 0, y = x^2$  and  $x + y = 2$  in the 1st quadrant

**Solution:** Points of intersection of  $y = x^2$  and  $x + y = 2$  are obtained as follows:

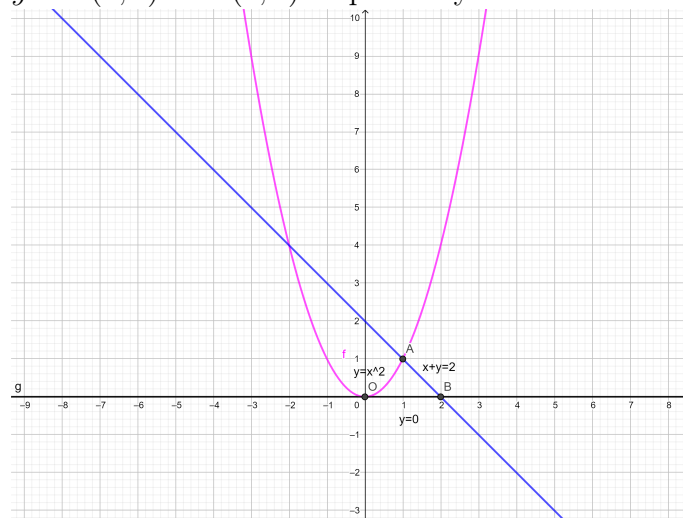
since,  $y = x^2$  and  $x + y = 2, \therefore x + x^2 = 2$

$\therefore x + x^2 - 2 = 0 \Rightarrow x = 1 \text{ or } x = -2$

if  $x = 1$ , then  $y = 1^2 = 1$ ; if  $x = -2$ , then  $y = (-2)^2 = 4$

Therefore, Points are  $(1, 1)$  and  $(-2, 4)$

Points of intersection of the line  $x + y = 2$  with co-ordinate axes  $x$  and  $y$  are  $(2, 0)$  and  $(0, 2)$  respectively.



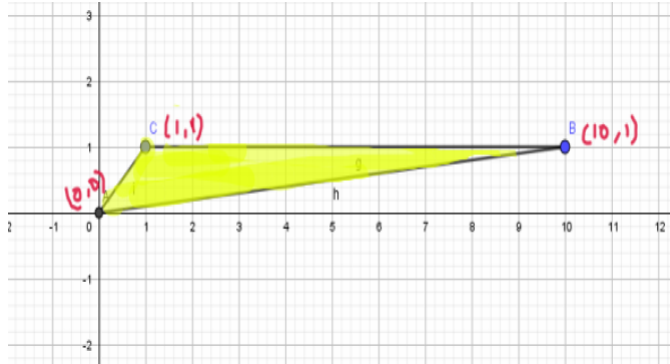
Now

$$\begin{aligned}\iint_{\text{given area}} y \, dx \, dy &= \int_{y=0}^{y=1} \left( \int_{x=\sqrt{y}}^{x=2-y} y \, dx \right) dy \\&= \int_{y=0}^{y=1} y[x] \Big|_{\sqrt{y}}^{2-y} dy \\&= \int_{y=0}^{y=1} y[2 - y - \sqrt{y}] dy \\&= \int_{y=0}^{y=1} [2y - y^2 - y^{3/2}] dy \\&= \left[ 2 \cdot \frac{y^2}{2} - \frac{y^3}{3} - \frac{y^{5/2}}{\frac{5}{2}} \right] \Big|_0^1 \\&= \left( 1 - \frac{1}{3} - \frac{2}{5} \right) - (0) \\&= \frac{2}{3} - \frac{2}{5} \\&= \frac{10 - 6}{15} \\&= \frac{4}{15}\end{aligned}$$

$$\text{or } \iint_{\text{given area}} y \, dx \, dy = \int_{x=0}^{x=1} \int_{y=0}^{y=x^2} y \, dx \, dy + \int_{x=1}^{x=2} \int_{y=0}^{y=2-x} y \, dx \, dy$$

4. Evaluate  $\iint_R \sqrt{xy - y^2} dx dy$  over a triangle whose vertices are  $(0,0)$ ,  $(10,1)$  and  $(1,1)$

Sol:



Equation of line OB is,

$$\frac{y-y_1}{y_1-y_2} = \frac{x-x_1}{x_1-x_2}$$

$$O \equiv (0, 0) \equiv (x_1, y_1)$$

$$B \equiv (10, 1) \equiv (x_2, y_2)$$

$$\therefore \frac{y-0}{0-1} = \frac{x-0}{0-10}$$

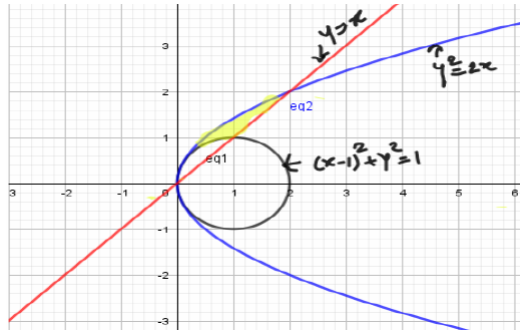
$$\therefore -y = \frac{-x}{10}$$

$$\therefore x = 10y$$

since line AB is parallel to x-axis passing through  $(0,1)$  therefore it's equation is  $y = 1$  since the line OA is passing through  $(0,0)$   $(1,1)$  therefore it's equation is  $y = x$

$$\begin{aligned} \iint_R \sqrt{xy - y^2} dx dy &= \int_{y=0}^{y=1} \left( \int_{x=y}^{x=10y} y^{1/2} (x - y)^{1/2} dx \right) dy \\ &= \int_{y=0}^{y=1} y^{1/2} \left[ \frac{(x-y)^{3/2}}{\frac{3}{2}} \right] \Big|_{x=y}^{x=10y} dy \\ &= \frac{2}{3} \int_{y=0}^{y=1} y^{1/2} \left[ (10y - y)^{3/2} - (y - y)^{3/2} \right] dy \\ &= \frac{2 \cdot 9^{3/2}}{3} \int_0^1 y^{1/2} y^{3/2} dy, \\ &= \frac{2 \cdot 3^3}{3} \int_0^1 y^2 dy \\ &= \frac{2}{3} \cdot 3^3 \cdot \frac{1}{3} = 6 \end{aligned}$$

5.  $\iint_R xy \, dx \, dy$  over the Region 'R' given by  $x^2 + y^2 - 2x = 0$  ,  $y^2 = 2x$   
 $y = x$   
 sol:



Equation  $x^2 + y^2 - 2x = 0$  can be written as  $(x - 1)^2 + y^2 = 1$  i.e a circle centered at (1,0) with radius '1'.

points of intersection of  $y = x$  and  $y^2 = 2x$  are given by ,  $x^2 = 2x$

$$x = 0 \text{ or } x = 2$$

$$y = 0 \mid y = 2$$

Points are (0,0) and (2,2)

and Point of intersection of  $y = x$  and  $x^2 + y^2 - 2x = 0$  are given by,

$$x^2 + y^2 - 2x = 0$$

$$\therefore 2x^2 - 2x = 0$$

$$x = 0 \text{ or } x = 1$$

$$y = 0 \mid y = 1$$

Points are (0,0) and (1,1)

and point of intersection of  $x^2 + y^2 - 2x = 0$  and  $y^2 = 2x$  is given by,

$$x^2 + y^2 - 2x = 0$$

$$\therefore x^2 = 0$$

$$\therefore x = 0$$

$$\therefore y = 0$$

$\therefore$  Point of intersection is (0,0)

entering boundary for line  $L_1$  is a circle  $x^2 + y^2 - 2x = 0$  which can be written as,  $y^2 = 2x - x^2$

$$\therefore y = \pm\sqrt{2x - x^2}$$



$\therefore y = \sqrt{2x - x^2}$  (since, Region is in the 1st quadrant)  
and Exist boundary is  $y^2 = 2x$  which is  $y = \pm\sqrt{2x} \therefore y = \sqrt{2x}$   
Similarly entering boundary for  $L_2$  is  $y = x$  and Exist boundary is  $y = \sqrt{2x}$

$$\begin{aligned}\therefore \text{Given Integral} &= \int_{x=0}^{x=1} \int_{y=\sqrt{2x-x^2}}^{y=\sqrt{2x}} xy \, dy \, dx + \int_{x=1}^{x=2} \int_{y=x}^{y=\sqrt{2x}} xy \, dy \, dx \\&= \int_{x=0}^{x=1} x \left[ \frac{y^2}{2} \right]_{y=\sqrt{2x-x^2}}^{y=\sqrt{2x}} dx + \int_{x=1}^{x=2} x \left[ \frac{y^2}{2} \right]_{y=x}^{y=\sqrt{2x}} dx \\&= \int_{x=0}^{x=1} x \left[ \frac{2x}{2} - \frac{2x-x^2}{2} \right] dx + \int_{x=1}^{x=2} x \left[ \frac{2x}{2} - \frac{x^2}{2} \right] dx \\&= \frac{1}{2} \int_0^1 (2x^2 - 2x^2 + x^3) dx + \frac{1}{2} \int_1^2 (2x^2 - x^3) dx \\&= \frac{1}{2} \left[ \frac{x^4}{4} \right]_0^1 + \frac{1}{2} \left[ 2 \cdot \frac{x^3}{3} - \frac{x^4}{4} \right]_1^2 \\&= \frac{1}{2} \left[ \frac{1}{4} - 0 \right] + \frac{1}{2} \left[ 2 \cdot \frac{2^3}{3} - \frac{2^4}{4} - \left( \frac{2}{3} - \frac{1}{4} \right) \right] \\&= \frac{1}{8} + \frac{1}{2} \left[ \frac{16}{3} - 4 - \frac{2}{3} + \frac{1}{4} \right] \\&= \frac{1}{8} + \frac{1}{2} \left[ \frac{14}{3} - \frac{15}{4} \right] \\&= \frac{1}{8} + \frac{1}{2} \cdot \frac{11}{12} \\&= \frac{1}{8} + \frac{11}{24} \\&= \frac{7}{12}\end{aligned}$$

## 4.2 Changing the order of Integration(No evaluation)

Sometimes evaluating the given integration in given order becomes difficult so changing the order of the integration may simplify the Evaluation.

Note:- Meaning of changing the order

Suppose given integration is,

$\int_{y=c}^{y=d} \int_{x=\phi_1(y)}^{x=\phi_2(y)} f(x, y) dx dy$  which is difficult to evaluate so by changing the order we can write given integral as,

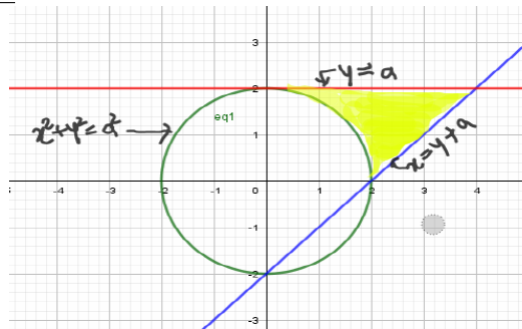
$\int_{x=a}^{x=b} \int_{y=g_1(x)}^{y=g_2(x)} f(x, y) dy dx$  which is may be easy to solve

first will see Examples on just changing the order of an Integration then examples on evaluation of integration by changing the order.

Q) Change the order of integration

1.  $\int_0^a \int_{\sqrt{a^2-y^2}}^{y+a} f(x, y) dx dy$  where 'a' is positive real number

sol:-



Given integration is,  $\int_{y=0}^{y=a} \int_{x=\sqrt{a^2-y^2}}^{x=y+a} f(x, y) dx dy$

since,  $x = \sqrt{a^2 - y^2}$  and  $x = y + a$

$\therefore x - y = a$  which is an equation of straight line.

$$\therefore x^2 = a^2 - y^2$$

$$\therefore x^2 + y^2 = a^2$$

which is an equation of a circle centered at origin with radius 'a'.

Point of intersection of a line  $x - y = a$  and  $y = a$  is given by,  $x - a = a$

$$\therefore x = 2a$$

$\therefore$  Point of intersection is  $(2a, a)$  since outer limits are from  $y = 0$  to  $y = a$  therefore region of Integration is present between the lines  $y = 0$  and  $y = a$ .

and since lower inner limit is  $x = \sqrt{a^2 - y^2}$  which represents a semi circle present in 1st and 4th quadrants [because  $x^2 + y^2 = a^2$  gives  $x^2 = a^2 - y^2$

$$\therefore x = \pm\sqrt{a^2 - y^2} \text{ but } x \text{ is given as } +\sqrt{a^2 - y^2}]$$

but region is present between  $y = 0$  and  $y = a$  therefore part of a circle present in 1st quadrant and upper inner limit is  $x = y + a$  which represents a straight line  $x - y = a$  therefore required region is bounded by part of a circle  $x^2 + y^2 = a^2$  present in 1st quadrant and the line  $x - y = a$  and lying between the lines  $y = 0$  and  $y = a$  as shown in figure below.

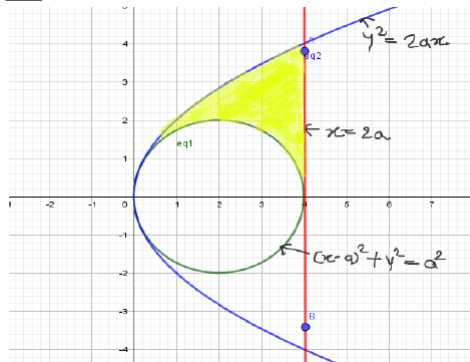
Now, in order to change the order of integration we need to pass lines parallel to y-axis.

Since entering boundaries are different therefore we need to split the region of integration .

$$\therefore \text{Required integral} = \int_{x=0}^{x=a} \int_{y=\sqrt{a^2-y^2}}^{y=a} f(x, y) dy dx + \int_{x=a}^{x=2a} \int_{y=x-a}^{y=a} f(x, y) dy dx$$

2. Change the order of  $\int_0^{2a} \int_{\sqrt{2ax-x^2}}^{\sqrt{2ax}} f(x, y) dy dx$  ( $a > 0$ )

sol:-



$$\text{Given integral} = \int_{x=0}^{x=2a} \int_{y=\sqrt{2ax-x^2}}^{y=\sqrt{2ax}} f(x, y) dy dx$$

$$\text{lower limit } y = \sqrt{2ax - x^2}$$

$$\therefore y^2 = 2ax - x^2$$

$$\therefore x^2 + y^2 - 2ax = 0$$

$\therefore (x - a)^2 + y^2 = a^2$  which is a circle centered at  $(a, 0)$  with radius 'a' .

upper limit  $y = \sqrt{2ax}$

$\therefore y^2 = 2ax$  which is a parabola symmetric to positive x-axis .

Required region is present between the lines  $x = 0$  and  $x = 2a$  and 'y' is changing from a part of a circle  $y = +\sqrt{2ax - x^2}$  present in 1st quadrant to a parabola  $y^2 = 2ax$  present in 1st quadrant as shown in the figure below.

Now we want to write inner limits as x-limits we need to pass line parallel to x-axis .But entering and exit boundaries are different for 1,2 and 3 regions therefore we need to split the integration.

we have  $(x - a)^2 + y^2 = a^2$

$$\therefore (x - a)^2 = a^2 - y^2$$

$$\therefore x - a = \pm\sqrt{a^2 - y^2} \rightarrow (1)$$

for 1,

$$0 \leq x \leq a$$

$$\therefore x - a \leq 0$$

$$\therefore x - a = -\sqrt{a^2 - y^2} \text{ from (1)}$$

$$\therefore x = a - \sqrt{a^2 - y^2}$$

for 2,

$$0 \leq x \leq 2a$$

$$\therefore x - a \geq 0$$

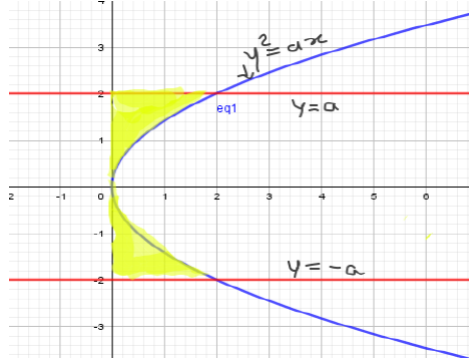
$$\therefore x - a = \sqrt{a^2 - y^2} \text{ from (1)}$$

$$\therefore x = a + \sqrt{a^2 - y^2}$$

$$\begin{aligned} \text{Required Integration} = & \int_{y=0}^{y=a} \int_{x=\frac{y^2}{2a}}^{x=a-\sqrt{a^2-y^2}} f(x, y) dx dy + \int_{y=a}^{y=2a} \int_{x=\frac{y^2}{2a}}^{x=2a} f(x, y) dx dy \\ & + \int_{y=0}^{y=a} \int_{x=a+\sqrt{a^2-y^2}}^{x=2a} f(x, y) dx dy \end{aligned}$$

3.  $\int_{-a}^a \int_0^{\frac{y^2}{a}} f(x, y) dx dy$  ( $a > 0$ )

sol:-



Given integral =  $\int_{y=-a}^{y=a} \int_{x=0}^{x=\frac{y^2}{a}} f(x, y) dx dy$

since outer limits are from  $y = -a$  to  $y = a$  therefore region is present between the lines  $y = -a$  and  $y = a$ .

and  $x = 0$  represents y-axis and  $x = \frac{y^2}{a}$  i.e.  $y^2 = ax$  represents a parabola symmetric to positive x-axis.

for 1,  $y^2 = ax \therefore y = \pm\sqrt{ax} \therefore y = -\sqrt{ax}$

(since Region is in 4th quadrant)

for 2,  $y^2 = ax \therefore y = \pm\sqrt{ax} \therefore y = +\sqrt{ax}$

(since Region is in 1st quadrant)

Required Integral =  $\int_{x=0}^{x=a} \int_{y=-\sqrt{ax}}^{y=+\sqrt{ax}} f(x, y) dy dx + \int_{x=0}^{x=a} \int_{y=-\sqrt{ax}}^{y=+\sqrt{ax}} f(x, y) dy dx$

4.  $\int_0^4 \int_{\frac{y}{2}}^{9-y} f(x, y) dx dy$

sol:-

Given integral =  $\int_{y=0}^{y=4} \int_{x=\frac{y}{2}}^{x=9-y} f(x, y) dx dy$

since outer limits are  $y = 0$  to  $y = 4$  therefore required region is present between the  $y = 0$  and  $y = 4$ .

$x = \frac{y}{2}$  gives  $y = 2x$  a straight line passing through origin

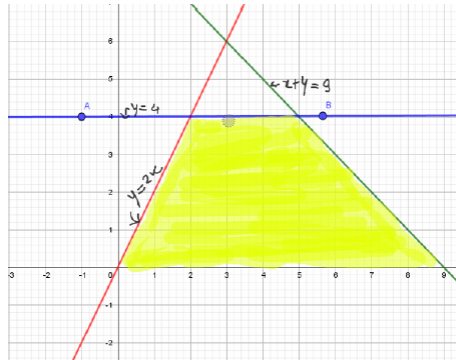
and  $x = 9 - y$  gives  $x + y = 9$  a straight line making an intercept of '9' on both the axes.

Point of intersection of  $y = 2x$  and  $x + y = 9$  is given as,  $x + 2x = 9$

$\therefore x = 3 \therefore y = 2(3) = 6$

$\therefore$  point is (3,6)

required region is shown below



(a) Point of intersection  $y = 4$  and  $y = 2x$  is obtained as,  $4 = 2x$

$\therefore x = 2$

$\therefore$  Point is (2,4)

(b) Point of intersection  $y = 4$  and  $x + y = 9$  is obtained as,  $x + y = 9$

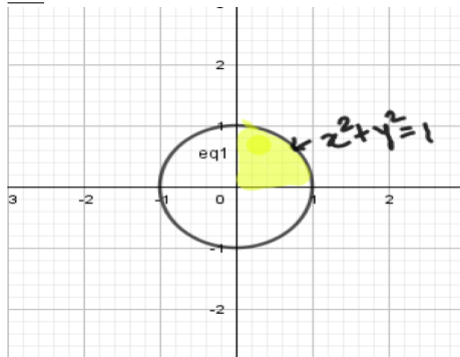
$\therefore x = 5$

$\therefore$  Point is (5,4)

5. Change the order of Integration

$$\int_0^1 \int_0^{\sqrt{1-x^2}} \frac{e^y}{(e^y+1)(\sqrt{1-x^2-y^2})} dy dx$$

sol:-



$$\text{Let } I = \int_{x=0}^1 \left( \int_{y=0}^{\sqrt{1-x^2}} \frac{e^y}{(e^y+1)(\sqrt{1-x^2-y^2})} dy \right) dx$$

'y' is changing from  $y=0$  (i.e x-axis) to  $y = \sqrt{1-x^2}$  (i.e  $y^2 = 1-x^2$ )

$\therefore x^2 + y^2 = 1$  which is a circle centre at (0,0) with radius '1' )

Note:-

- (a) Here integrating w.r.t. 'dy' is difficult as numerator is a function of 'y' denominator is a product of two function of 'y' in which one is Exponential other is inside the square root but if we integrate w.r.t 'dx' then most of terms become constant go outside integration so integration becomes easy.
- (b) But mostly in the Examination it will be given in the question that change the order Evaluate.

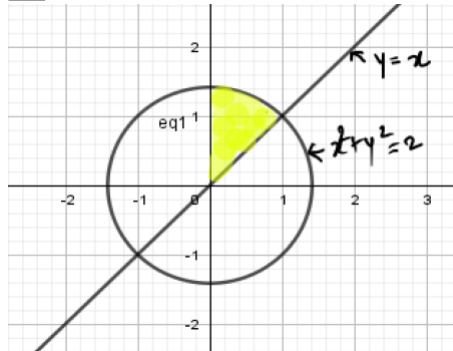
So after changing the order we have,

$$I = \int_{y=0}^{y=1} \left( \int_{x=0}^{x=\sqrt{1-y^2}} \frac{e^y}{(e^y+1)(\sqrt{1-x^2-y^2})} dy \right) dx$$

#### 6. Change the order of Integration

$$\int_0^1 dx \int_x^{\sqrt{2-x^2}} \frac{x}{\sqrt{x^2+y^2}} dy$$

sol:



$$\text{Let } I = \int_0^1 dx \int_{y=x}^{y=\sqrt{2-x^2}} \frac{x}{\sqrt{x^2+y^2}} dy$$

'y' is changing from  $y = x$  (a straight line passing through origin) to  $y = \sqrt{2-x^2}$  (i.e  $y^2 = 2 - x^2 \therefore x^2 + y^2 = (\sqrt{2})^2$ )

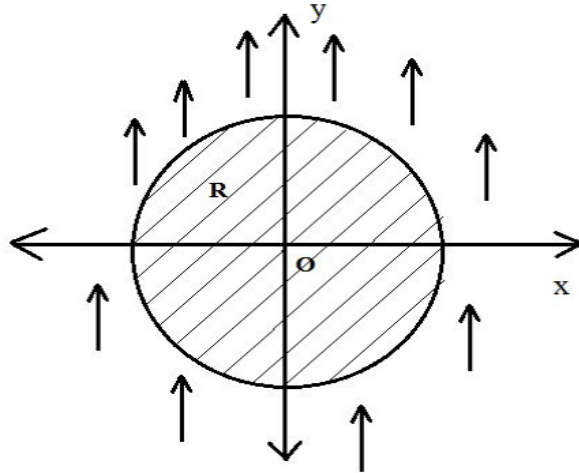
After changing the order we have,

$$I = \int_{y=0}^{y=1} \left( \int_{x=0}^{x=y} \frac{x}{\sqrt{x^2+y^2}} dx \right) dy + \int_{y=1}^{y=\sqrt{2}} \left( \int_{x=0}^{x=\sqrt{2-y^2}} \frac{x}{\sqrt{x^2+y^2}} dx \right) dy$$

### 4.3 Polar Coordinates

Consider the integral  $\int \int_R e^{-(x^2+y^2)} dx dy$ , where  $R$  is the region  $x^2 + y^2 \leq 1$

Note:-  $x^2 + y^2 \leq 1$  represents a disc centered at origin with radius 1.



On boundary  $x^2 + y^2 = 1 \implies y^2 = 1 - x^2 \implies y = \pm\sqrt{1 - x^2}$

For 1st and 2nd quadrants:  $y = +\sqrt{1 - x^2}$

and for 3rd and 4th quadrants  $y = -\sqrt{1 - x^2}$

$\therefore$  Given Integral =  $\int_{x=-1}^{x=1} \int_{y=-\sqrt{1-x^2}}^{y=+\sqrt{1-x^2}} e^{-(x^2+y^2)} dy dx$

which is difficult to solve (but not impossible) as limits contain square root of  $(1 - x^2)$  and the function to be integrated is difficult. So let's convert this integral into polar coordinates and let's see what happens.

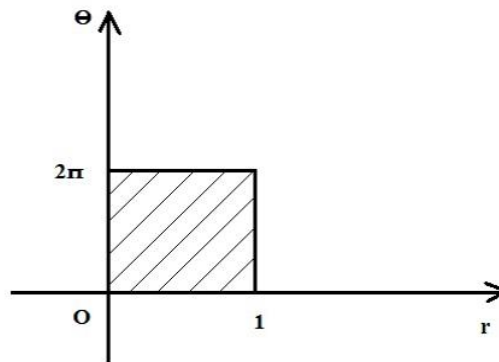
As we know that for polar coordinates  $x = r \cos \theta$  and  $y = r \sin \theta$

so for the given Region  $R$  in  $xy$  plane we have  $0 \leq r \leq 1$  and  $0 \leq \theta \leq 2\pi$

**Note:-** How to find  $r$  and  $\theta$  limits will be explained later.

So given region in  $xy$  plane get transform into a region bounded by rectangle in  $r - \theta$  plane.





and integrating over region bounded by rectangle is easier (Because limits are constants)

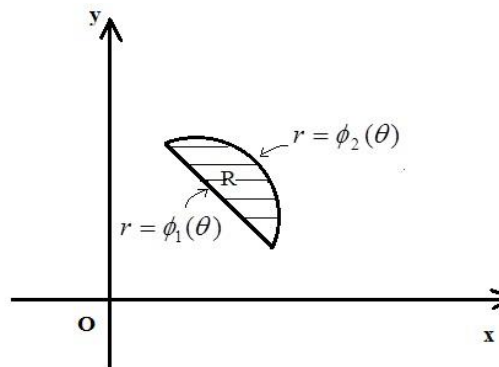
$$\therefore \text{Given Integral} = \int_{\theta=0}^{\theta=2\pi} \int_{r=0}^{r=1} e^{-r^2} r dr d\theta$$

**Note:-** When we convert into polar coordinates we replace  $dx dy$  by  $r dr d\theta$

### Finding limits in Polar Coordinates

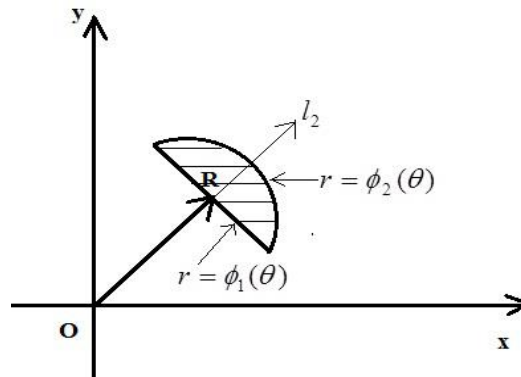
First note that for finding ' $r$ ' and ' $\theta$ ' limits we will be using region of  $xy$  plane [i.e we will not be converting the given region into  $r - \theta$  plane]. For integrating first with respect to ' $r$ ' and then with respect to ' $\theta$ ' we take the following steps:

- (a) Sketch the region and label the boundary curves.



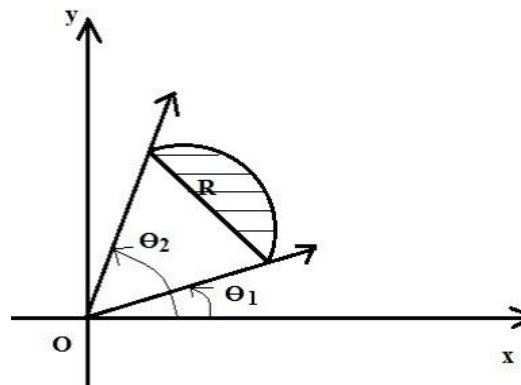
Express boundary curves in terms of ' $r$ ' and ' $\theta$ ' by using the substitution  $x = r \cos \theta$  and  $y = r \sin \theta$

- (b) To find ' $r$ ' limits imagine a ray ' $l$ ' from the origin cutting through ' $R$ ' in the direction of increasing ' $r$ ' mark the  $r$  - values where ' $l$ ' enters leaves  $R$  these are the  $r$ -limits of integration they usually depend on ' $\theta$ ' some-times constant.



**Note:-** 'r' limits are always positive.

(c) To find  $\theta$ -limits: find the smallest and largest  $\theta$ -values that bound 'R' and these are the  $\theta$ -limits of integration.



### Change of variables in Double integration

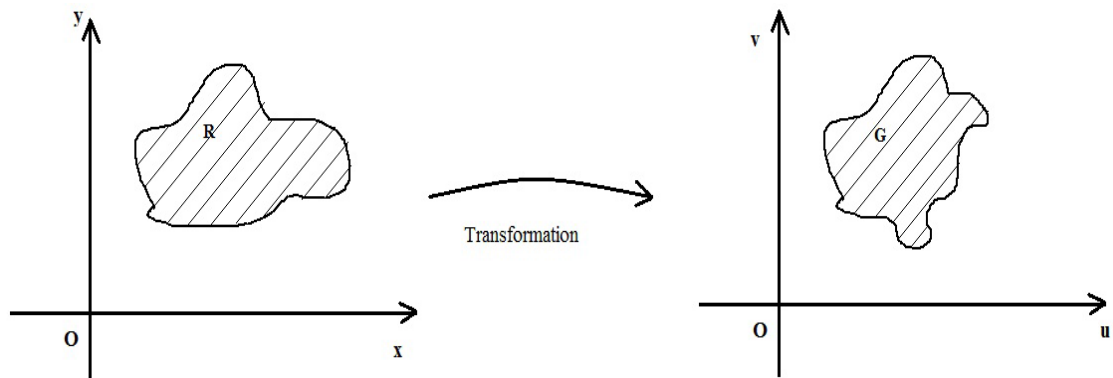
We have already seen that in single integration  $\int_a^b f(x)dx$ , for solving some complicated integrations we put  $x = \phi(t)$  then we replace as,  $dx = \phi'(t)dt$  and we change the limits accordingly so we get

$$\int_a^b f(x)dx = \int_c^d f(\phi(t)) \cdot \phi'(t)dt$$

where 'c' and 'd' are new limits.

Similarly in double integration, when we change a problem in xy-plane to some other u-v plane by substituting  $x = \phi_1(u, v)$  and  $y = \phi_2(u, v)$  we get,

$$\int \int_R f(x, y) dx dy = \int \int_G g(u, v) |J| du dv$$



where  $|J|$  = absolute value of Jacobian and Jacobian is defined as  $J =$

$$\begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix} \quad [\text{Jacobian defines differentiability in } xy\text{-plane}] \quad \textbf{Note:-} \text{ But in this}$$

semester most of the time we will be using polar coordinates i.e  $x = r \cos \theta$  and  $y = r \sin \theta$  for simplifying the calculations

Jacobian in this case is,

$$J = \begin{vmatrix} x_r & y_r \\ x_\theta & y_\theta \end{vmatrix} = \begin{vmatrix} \cos \theta & \sin \theta \\ -r \sin \theta & r \cos \theta \end{vmatrix} = r \cos^2 \theta + r \sin^2 \theta = r$$

Therefore

$$\int \int_R f(x, y) \, dx \, dy = \int \int_G f(r \cos \theta, r \sin \theta) \, r \, dr \, d\theta$$

**Note:-** Three types of example expected on polar coordinates

- Evaluate the integration by finding ' $r$ ' and ' $\theta$ ' limits.
- Integration will be given in terms of  $x$  and  $y$  and you will be asked to convert in terms of ' $r$ ' and ' $\theta$ ' do the Evaluation.
- We need to understand that the given problem should be converted in terms of  $r$  and  $\theta$  [in order to simplify the integration ] and do the evaluation.

### Solved Examples:

1. Solve  $\int \int_R r^3 dr d\theta$ , where  $R$  is bounded by the circles  $r = 2 \sin \theta$  and  $r = 4 \sin \theta$ .

Solution: First sketch the region of Integration for that we need to identify the curves by converting in  $x$  and  $y$ .

$$r = 2 \sin \theta \implies r^2 = 2r \sin \theta \implies x^2 + y^2 = 2y \quad [\because x = r \cos \theta \text{ \& } y = r \sin \theta]$$

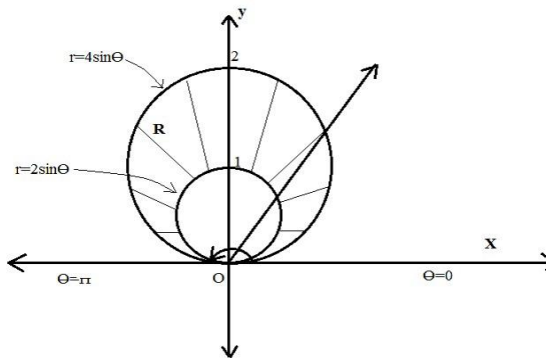
$$x^2 + y^2 - 2y = 0 \implies x^2 + y^2 - 2y + 1 = 1 \implies x^2 + (y - 1)^2 = 1$$

which is a circle centered at  $(0, 1)$  with radius 1.

$$r = 4 \sin \theta \implies r^2 = 4r \sin \theta \implies x^2 + y^2 = 4y \quad [\because x = r \cos \theta \text{ \& } y = r \sin \theta]$$

$$x^2 + y^2 - 4y = 0 \implies x^2 + y^2 - 2y + 4 = 4 \implies x^2 + (y - 2)^2 = 4$$

which is a circle centered at  $(0, 2)$  with radius 2.



$$\begin{aligned} \therefore I &= \int \int_R r^3 dr d\theta = \int_{\theta=0}^{\theta=\pi} \int_{r=2 \sin \theta}^{r=4 \sin \theta} r^3 dr d\theta \\ &= \int_{\theta=0}^{\theta=\pi} \left. \frac{r^4}{4} \right|_{r=2 \sin \theta}^{r=4 \sin \theta} d\theta \\ &= \frac{1}{4} \int_{\theta=0}^{\theta=\pi} [4^4 \sin^4 \theta - 2^4 \sin^4 \theta] d\theta \\ &= \frac{4^4 - 2^4}{4} \int_0^\pi \sin^4 \theta d\theta = 60 \int_0^\pi \left( 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} \right)^4 d\theta \\ &= 60 \cdot 2^4 \int_0^\pi \sin^4 \frac{\theta}{2} \cdot \cos^4 \frac{\theta}{2} d\theta \end{aligned}$$

$$\text{Put } x = \frac{\theta}{2} \implies dx = \frac{1}{2}d\theta \quad \text{and} \quad \theta : 0 \rightarrow \pi \implies x : 0 \rightarrow \frac{\pi}{2}$$

$$\begin{aligned} \therefore I &= 960 \int_0^{\pi/2} \sin^4 x \cdot \cos^4 x \, 2dx \\ &= 1920 \int_0^{\pi/2} \sin^4 x \cdot \cos^4 x \, dx \\ &= 1920 \frac{1}{2} \beta \left( \frac{4+1}{2}, \frac{4+1}{2} \right) \left[ \int_0^{\pi/2} \sin^p \theta \cdot \cos^q \theta \, d\theta = \frac{1}{2} \beta \left( \frac{p+1}{2}, \frac{q+1}{2} \right) \right] \\ &= 960 \beta \left( \frac{5}{2}, \frac{5}{2} \right) = 960 \frac{\Gamma \left( \frac{5}{2} \right) \Gamma \left( \frac{5}{2} \right)}{\Gamma \left( \frac{5}{2} + \frac{5}{2} \right)} = 960 \frac{\Gamma \left( \frac{5}{2} \right) \Gamma \left( \frac{5}{2} \right)}{\Gamma(5)} \\ &= 960 \frac{\frac{3}{2} \cdot \frac{1}{2} \Gamma \left( \frac{1}{2} \right) \cdot \frac{3}{2} \cdot \frac{1}{2} \Gamma \left( \frac{1}{2} \right)}{4!} = 960 \frac{3\sqrt{\pi} \cdot 3\sqrt{\pi}}{4! \cdot 2^4} = 15 \frac{3\pi}{2} \end{aligned}$$

$$\therefore I = \frac{45\pi}{2}$$

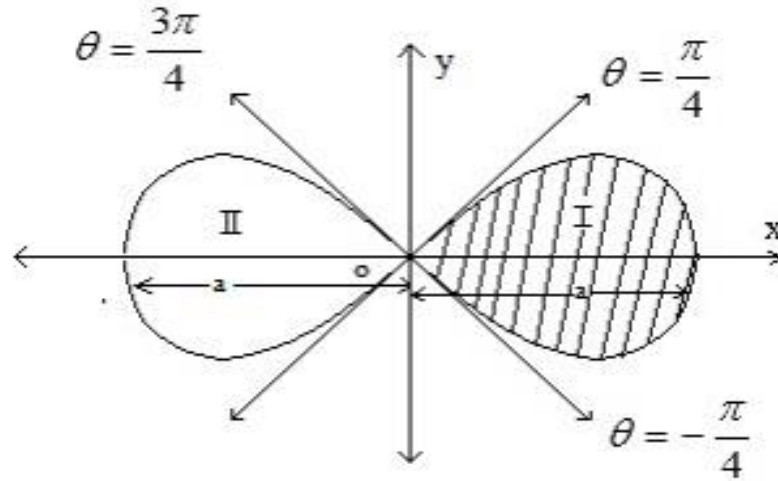
2. Evaluate  $\int \int \frac{r \, dr \, d\theta}{\sqrt{r^2 + a^2}}$  over one loop of lemniscate  $r^2 = a^2 \cos 2\theta$ .

Solution: First we sketch the lemniscate  $r^2 = a^2 \cos 2\theta$  [Cartesian form is  $(x^2 + y^2)^2 = a^2(x^2 - y^2)$ ]

- (a) On changing 'r' by '-r' equation remain unchanged therefore the curve is symmetric about origin.
- (b) On changing ' $\theta$ ' by ' $-\theta$ ' equation remain unchanged therefore the curve is symmetric about positive x-axis.

(c)

|          |     |                 |                                           |                  |       |
|----------|-----|-----------------|-------------------------------------------|------------------|-------|
| $\theta$ | 0   | $\frac{\pi}{4}$ | $\frac{\pi}{4} < \theta < \frac{3\pi}{4}$ | $\frac{3\pi}{4}$ | $\pi$ |
| $r$      | $a$ | 0               | no part of the curve since $r^2 < 0$      | 0                | $a$   |



$$\begin{aligned}
 \therefore I &= \int_{\theta=-\frac{\pi}{4}}^{\theta=\frac{\pi}{4}} \int_{r=0}^{r=a\sqrt{\cos 2\theta}} \frac{r \, dr \, d\theta}{\sqrt{r^2 + a^2}} \\
 &= \int_{\theta=-\frac{\pi}{4}}^{\theta=\frac{\pi}{4}} \left( \int_{r=0}^{r=a\sqrt{\cos 2\theta}} \frac{r}{\sqrt{r^2 + a^2}} \, dr \right) d\theta \\
 &= \int_{\theta=-\frac{\pi}{4}}^{\theta=\frac{\pi}{4}} \left( \sqrt{r^2 + a^2} \right) \Big|_{r=0}^{r=a\sqrt{\cos 2\theta}} d\theta \\
 &= \int_{\theta=-\frac{\pi}{4}}^{\theta=\frac{\pi}{4}} \left( \sqrt{a^2 \cos 2\theta + a^2} - a \right) d\theta \\
 &= a \int_{\theta=-\frac{\pi}{4}}^{\theta=\frac{\pi}{4}} \left( \sqrt{\cos 2\theta + 1} - 1 \right) d\theta \\
 &= a \int_{\theta=-\frac{\pi}{4}}^{\theta=\frac{\pi}{4}} \left( \sqrt{2} \cos \theta - 1 \right) d\theta \\
 &= a \left( \sqrt{2} \sin \theta - \theta \right) \Big|_{\theta=-\frac{\pi}{4}}^{\theta=\frac{\pi}{4}} \\
 &= a \left( \sqrt{2} \left( \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right) - \left( \frac{\pi}{4} + \frac{\pi}{4} \right) \right) \\
 \therefore I &= \frac{a}{2} (4 - \pi)
 \end{aligned}$$

3. Express in polar co-ordinates and Evaluate

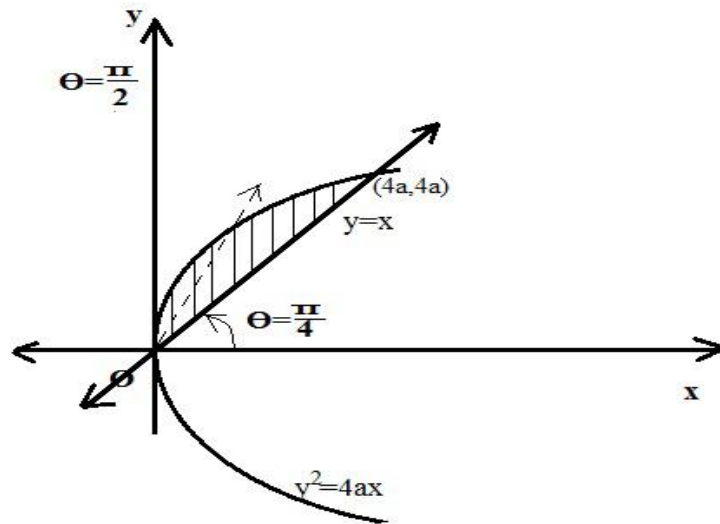
$$\int_0^{4a} \int_{\frac{y^2}{4a}}^y \frac{x^2 - y^2}{x^2 + y^2} dx dy \quad (a > 0)$$

Solution: Let  $I = \int_{y=0}^{y=4a} \int_{x=\frac{y^2}{4a}}^{x=y} \frac{x^2 - y^2}{x^2 + y^2} dx dy$

Points of intersection of  $x = y$  and  $x = \frac{y^2}{4a}$

$$\therefore y = \frac{y^2}{4a} \implies y^2 = 4ay \implies y = 0 \text{ or } y = 4a$$

Therefore points are  $(0,0)$  and  $(4a, 4a)$



'r' is changing from '0' to parabola  $y^2 = 4ax$  is  $r^2 \sin^2 \theta = 4ar \cos \theta$   
 $\therefore r = \frac{4a \cos \theta}{\sin^2 \theta}$

$$\begin{aligned} \therefore I &= \int_{\theta=\frac{\pi}{4}}^{\theta=\frac{\pi}{2}} \int_{r=0}^{r=\frac{4a \cos \theta}{\sin^2 \theta}} \left( \frac{r^2 \cos^2 \theta - r^2 \sin^2 \theta}{r^2 \cos^2 \theta + r^2 \sin^2 \theta} \right) r dr d\theta \\ &= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\cos^2 \theta - \sin^2 \theta) \left( \int_0^{\frac{4a \cos \theta}{\sin^2 \theta}} r dr \right) d\theta \\ &= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\cos^2 \theta - \sin^2 \theta) \left[ \frac{r^2}{2} \right]_0^{\frac{4a \cos \theta}{\sin^2 \theta}} d\theta \\ &= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\cos^2 \theta - \sin^2 \theta) \frac{8a^2 \cos^2 \theta}{\sin^4 \theta} d\theta \\ &= 8a^2 \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\cot^4 \theta - \cot^2 \theta) d\theta \end{aligned}$$

$$\begin{aligned}
 \therefore I &= \int_{\theta=\frac{\pi}{4}}^{\frac{\pi}{2}} \int_{r=0}^{r=\frac{4a \cos \theta}{\sin^2 \theta}} \left( \frac{r^2 \cos^2 \theta - r^2 \sin^2 \theta}{r^2 \cos^2 \theta + r^2 \sin^2 \theta} \right) r \, dr \, d\theta \\
 &= 8a^2 \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^2 \theta (\operatorname{cosec}^2 \theta - 1 - 1) \, d\theta \\
 &= 8a^2 \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\cot^2 \theta \operatorname{cosec}^2 \theta - 2 \cot^2 \theta) \, d\theta \\
 &= 8a^2 \left[ \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^2 \theta \operatorname{cosec}^2 \theta \, d\theta - 2 \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \operatorname{cosec}^2 \theta \, d\theta + 2 \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} d\theta \right] \\
 &= 8a^2 \left[ \frac{-1}{3} \cot^3 \theta - 2(-\cot \theta) + 2\theta \right]_{\frac{\pi}{4}}^{\frac{\pi}{2}} \\
 &= 8a^2 \left[ \frac{1}{3} - 2 + 2\frac{\pi}{4} \right] \\
 \therefore I &= 8a^2 \left[ \frac{\pi}{2} - \frac{5}{3} \right]
 \end{aligned}$$

4. Evaluate  $\int_0^a \int_{\sqrt{ax-x^2}}^{\sqrt{a^2-x^2}} \frac{dx \, dy}{\sqrt{a^2-x^2-y^2}}$  by converting in polar coordinates.

Solution: Let  $I = \int_{x=0}^{x=a} \int_{y=\sqrt{ax-x^2}}^{y=\sqrt{a^2-x^2}} \frac{dy}{\sqrt{a^2-x^2-y^2}} \, dx$

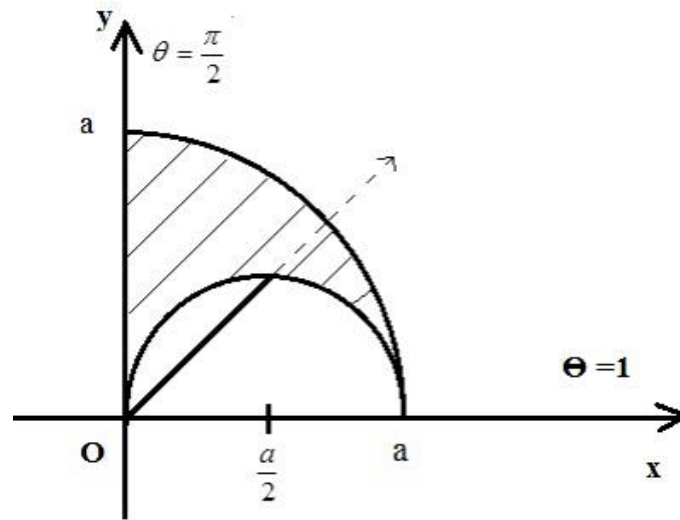
$$y = \sqrt{ax-x^2} \implies y^2 = ax-x^2 \implies \left(x - \frac{a}{2}\right)^2 + y^2 = \left(\frac{a}{2}\right)^2$$

which is a circle centered at  $\left(\frac{a}{2}, 0\right)$  with radius  $\frac{a}{2}$ .

$$\text{and } y = \sqrt{a^2-x^2} \implies y^2 = a^2-x^2 \implies x^2+y^2 = a^2$$

which is a circle centered at  $(0, 0)$  with radius  $a$ .





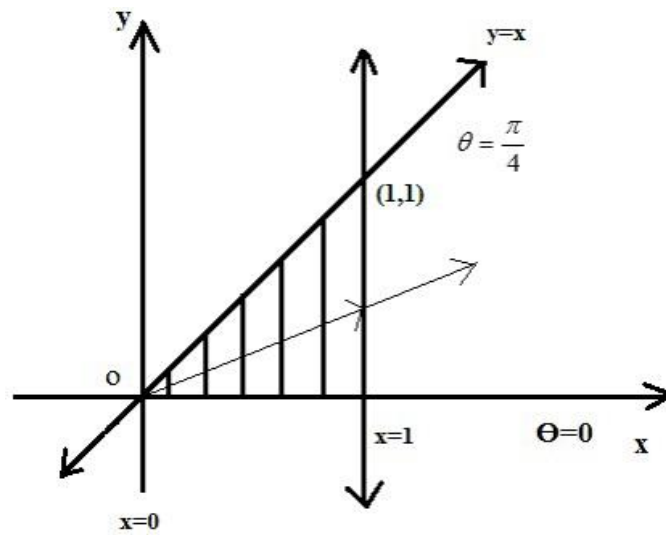
For inner circle:  $x^2 + y^2 = ax \implies r^2 = ar \cos \theta \implies r = a \cos \theta$

For outer circle:  $x^2 + y^2 = a^2 \implies r^2 = a^2 \implies r = a$

$$\begin{aligned} \therefore I &= \int_0^{\frac{\pi}{2}} \int_{r=a \cos \theta}^{r=a} \frac{r \, dr \, d\theta}{\sqrt{a^2 - r^2}} = \int_0^{\frac{\pi}{2}} -\sqrt{a^2 - r^2} \Big|_{r=a \cos \theta}^{r=a} d\theta \\ &= \int_0^{\frac{\pi}{2}} (0 + a \sin \theta) \, d\theta \\ &= -a \cos \theta \Big|_0^{\frac{\pi}{2}} \\ \therefore I &= a \end{aligned}$$

5. Evaluate  $\int_0^1 \int_0^x (x + y) \, dx \, dy$  by converting into polar coordinates.

Solution: Let  $I = \int_{x=0}^{x=1} \int_{y=0}^{y=x} (x + y) \, dx \, dy$

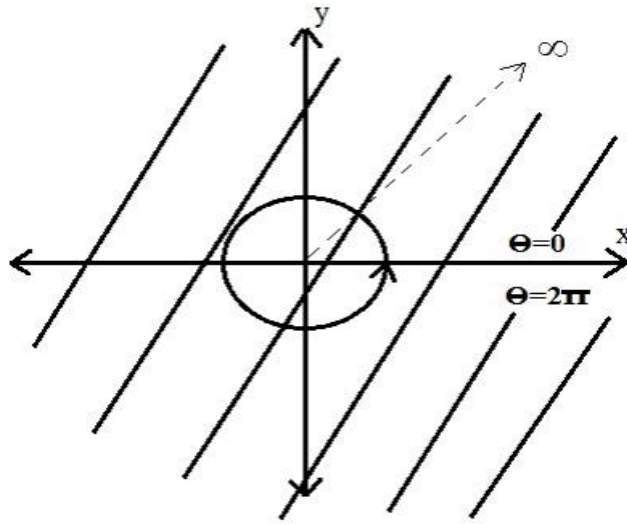


' $r$ ' will change from '0' to the line  $x = 1$  which is  $r \cos \theta = 1 \implies r = \sec \theta$

$$\begin{aligned}
 \therefore I &= \int_0^{\frac{\pi}{4}} \int_{r=0}^{r=\sec \theta} [r \cos \theta + r \sin \theta] r \, dr \, d\theta \\
 &= \int_0^{\frac{\pi}{4}} (\cos \theta + \sin \theta) \left[ \frac{r^3}{3} \right]_{r=0}^{r=\sec \theta} d\theta \\
 &= \frac{1}{3} \int_0^{\frac{\pi}{4}} (\cos \theta + \sin \theta) \sec^3 \theta \, d\theta \\
 &= \frac{1}{3} \int_0^{\frac{\pi}{4}} (\sec^2 \theta + \tan \theta \sec^2 \theta) \, d\theta \\
 \therefore I &= \frac{1}{2}
 \end{aligned}$$

6. Evaluate  $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{dx \, dy}{(a^2 + x^2 + y^2)^{3/2}}$  by converting into polar coordinates.

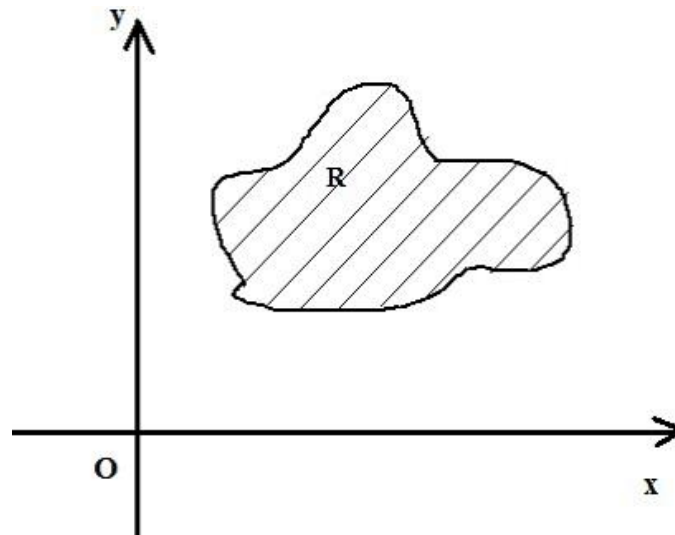
Solution: Let  $I = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{dx \, dy}{(a^2 + x^2 + y^2)^{3/2}}$



$$\begin{aligned}
 \therefore I &= \int_0^{2\pi} \int_{r=0}^{r=\infty} \frac{r \, dr \, d\theta}{(a^2 + r^2)^{3/2}} \\
 &= \int_0^{2\pi} \left. -(a^2 + r^2)^{-1/2} \right|_{r=0}^{r=\infty} d\theta \\
 &= \int_0^{2\pi} \left( -0 + \frac{1}{a} \right) d\theta \\
 &= \frac{1}{a} \int_0^{2\pi} d\theta \\
 &= \frac{1}{a} [\theta]_0^{2\pi} \\
 \therefore I &= \frac{2\pi}{a}
 \end{aligned}$$

#### 4.4 Applications of Double Integrals

**Area:** If we take  $f(x, y) = 1$  in the definition of the double integral over a region 'R', we get Area of the region 'R'.



$$\text{Area of Region 'R'} = \int \int_R 1 \, dx \, dy$$

##### Solved Examples:

- Find the area between the parabola  $y = x^2 - 6x + 3$  and the line  $y = 2x - 9$ .

Solution: First sketch the region of Integration.

$$y = x^2 - 6x + 3 \implies y = x^2 - 6x + 9 - 6 \implies y = (x - 3)^2 - 6$$

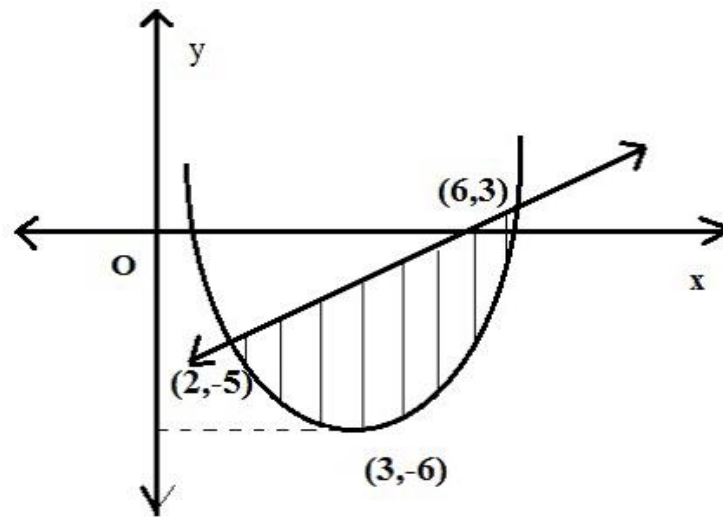
$\therefore y - (-6) = (x - 3)^2$ , which is a parabola facing upwards and having vertex at  $(3, -6)$

Now to find points of intersection of  $y = x^2 - 6x + 3$  and  $y = 2x - 9$

$$\therefore x^2 - 6x + 3 = 2x - 9 \implies x^2 - 8x + 12 = 0 \implies x = 2 \text{ or } x = 6$$

and  $y = -5$  or  $y = 3$

Therefore, Points are  $(2, -5)$  and  $(6, 3)$ .



$$\begin{aligned}
 \therefore A &= \int_{x=2}^{x=6} \int_{y=x^2-6x+3}^{y=2x-9} 1 \, dy \, dx \\
 &= \int_{x=2}^{x=6} y \Big|_{y=x^2-6x+3}^{y=2x-9} dx \\
 &= \int_{x=2}^{x=6} (2x - 9 - x^2 + 6x - 3) \, dx \\
 &= \int_{x=2}^{x=6} (8x - x^2 - 12) \, dx \\
 &= \left[ 4x^2 - \frac{x^3}{3} - 12x \right]_2^6 \\
 &= 4(6^2 - 2^2) - \frac{6^3 - 2^3}{3} - 12(6 - 2) \\
 \therefore A &= \frac{32}{3}
 \end{aligned}$$

2. Find the area bounded by the curve  $y^2(2a-x) = x^3$  and its asymptote..

Solution: First we sketch the Region

$$y^2(2a-x) = x^3 \implies y = \pm \sqrt{\frac{x^3}{2a-x}}$$

So as  $x \rightarrow 2a \implies y \rightarrow \pm\infty$

If  $x < 0$ ,  $y^2$  is negative which is not possible.

Therefore no part of the curve to the left side of y-axis.

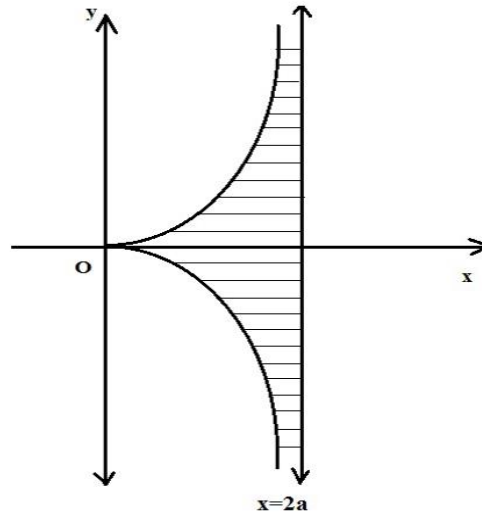
**Note:-** Asymptote of the graph  $y = f(x)$  is a line  $x = p$  (or  $y = p$ )

such that

$$\lim_{x \rightarrow p^+} f(x) = \pm\infty$$

or

$$\lim_{x \rightarrow p^-} f(x) = \pm\infty$$



$$\begin{aligned} \therefore A &= 2 \int_0^{2a} \int_{y=0}^{y=\sqrt{\frac{x^3}{2a-x}}} 1 \, dy \, dx \\ &= 2 \int_0^{2a} y \Big|_{y=0}^{y=\sqrt{\frac{x^3}{2a-x}}} dx \\ &= 2 \int_0^{2a} \sqrt{\frac{x^3}{2a-x}} dx \end{aligned}$$

now put  $x = 2at$  and solve [Hint: use Beta and Gamma functions]

$$\therefore A = 3\pi a^2$$

**Note:-** i) It is an infinite region but it has finite Area (approx.)

ii) **Area formula in polar coordinates** is  $\text{Area} = \int \int_R r \, dr \, d\theta$ .

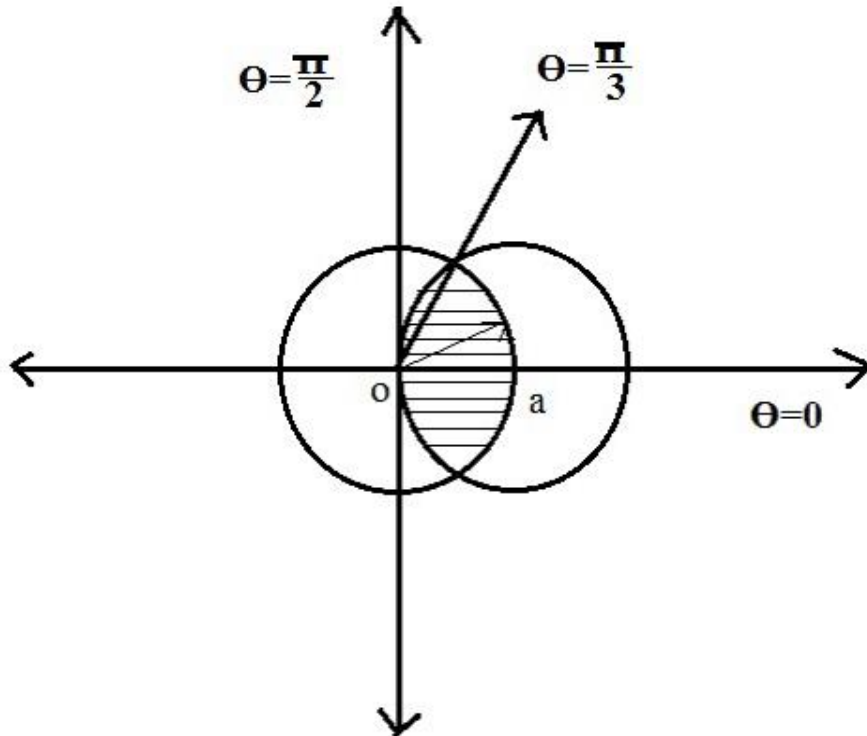
3. Find the area common of the circles  $r = a$  and  $r = 2a \cos \theta$ .

Solution:  $r = a \implies x^2 + y^2 = a^2$ , circle with centre at  $(0, 0)$  and radius ' $a$ '

$$\text{and } r = 2a \cos \theta \implies r^2 = 2ar \cos \theta \implies x^2 + y^2 = 2ax \implies (x - a)^2 + y^2 = a^2$$

circle with centre at  $(a, 0)$  and radius ' $a$ '

$$\text{Two circles intersect at } a = 2a \cos \theta \implies \cos \theta = \frac{1}{2} \implies \therefore \theta = \frac{\pi}{3}$$

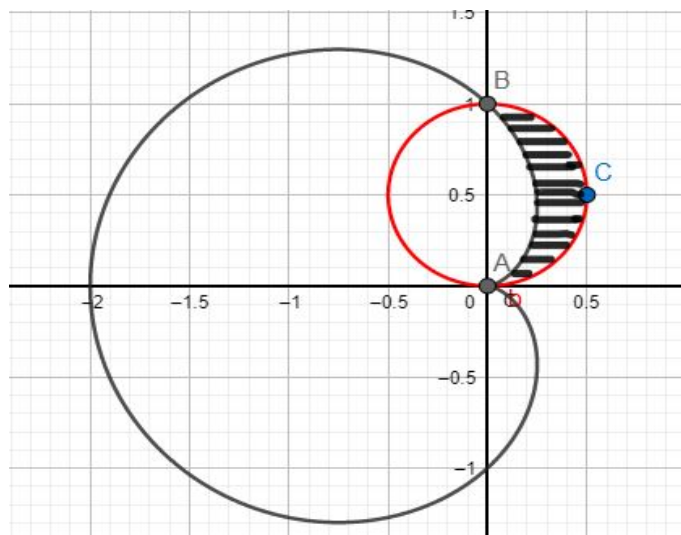


$$\begin{aligned} \therefore A &= 2 \left[ \int_{\theta=0}^{\theta=\frac{\pi}{3}} \int_{r=0}^{r=a} r \, dr \, d\theta + \int_{\theta=\frac{\pi}{3}}^{\theta=\frac{\pi}{2}} \int_{r=0}^{r=a \cos \theta} r \, dr \, d\theta \right] \\ &= \frac{2\pi a^2}{3} - \frac{\sqrt{3}a^2}{2} \end{aligned}$$

4. Find the area inside the circle  $r = a \sin \theta$  and outside the cardioid  $r = a(1 - \cos \theta)$ .

Solution: Let's plot cardioid  $r = a(1 - \cos \theta)$ .

|          |   |                 |       |                  |        |
|----------|---|-----------------|-------|------------------|--------|
| $\theta$ | 0 | $\frac{\pi}{2}$ | $\pi$ | $\frac{3\pi}{2}$ | $2\pi$ |
| $r$      | 0 | $a$             | $2a$  | $a$              | 0      |



$$\therefore A = \int_0^{\frac{\pi}{2}} \int_{r=a(1-\cos\theta)}^{r=a\sin\theta} r \, dr \, d\theta$$

$$\therefore A = \frac{a^2(2-\pi)}{2}$$